Project: Worked examples
Title: Psychrometrics
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

The state of moist air (humidity ratio, relative humidity, dew point, wet-bulb, enthalpy, specific volume, density) from dry-bulb and one other property at a stated pressure or altitude, and one air-handling process from that state with its heat and moisture balance: sensible heating or cooling, cooling with dehumidification, steam humidification, adiabatic humidification, or mixing with a second stream. $p_{ws}$ from Hyland–Wexler (over water ≥ 0 °C, over ice below); $W = 0.621945\, p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$; $h = 1.006\, t + W(2501 + 1.86\, t)$ kJ/kg_da; $v = 0.287042\,(t + 273.15)(1 + 1.607858\, W)/p$; dew point and wet-bulb by bisection (wet-bulb from the adiabatic-saturation relation). Processes on a dry-air mass flow: sensible $Q = \dot{m}\,(h_2 - h_1)$ at constant W; cooling/dehumidification $Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\, h_w$ with condensate $\dot{m}_w = \dot{m}(W_1 - W_2)$; steam $h_2 = h_1 + (W_2 - W_1)\, h_{steam}$; adiabatic humidification at constant h; mixing by mass-weighted W and h.

Ref: ASHRAE Handbook — Fundamentals Ch. 1 — Psychrometrics (ideal-gas relations, processes)
Ref: Hyland & Wexler, ASHRAE Trans. 89(2A), 1983 — Saturation pressure of water and ice
Ref: CIBSE Guide C Section 1 — Properties of humid air (principles)

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| altitude | Altitude (used when pressure is blank) | 0 | m | assumption: Sea-level site. | -500 … 5000 m |
| t_db | Dry-bulb temperature | 28 | degC | spec: CIBSE Guide A summer design condition for the site, assumed | -40 … 60 degC |
| known | Second known property | relative humidity | | assumption: RH given. | any |
| RH | Relative humidity | 60 | | spec: Design condition | 0 … 100 |
| m_da | Dry-air mass flow | 2.4 | kg/s | spec: AHU schedule AHU-01: 2.0 m³/s ≈ 2.4 kg/s dry air | 0 … 1000 kg/s |
| process | Process | cooling with dehumidification | | assumption: Chilled-water coil. | any |
| t_2 | Leaving dry-bulb (sensible, cooling) | 12 | degC | spec: AHU schedule AHU-01 off-coil 12 °C | -40 … 80 degC |
| RH_2 | Leaving relative humidity (cooling, evaporative) | 95 | | assumption: Off-coil 95 % RH (typical contact factor). | 0 … 100 |
| h_steam | Steam enthalpy (steam humidification) | 2676 | | default | 2500 … 3000 |

## 3. Assumptions

A1. Moist air is an ideal mixture of ideal gases; the enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. Condensate leaves at the coil leaving dry-bulb; steam adds enthalpy h_steam at negligible mass of dry air. (Process conventions.)
A3. Adiabatic humidification follows a line of constant enthalpy (close to constant wet-bulb). (Evaporative process idealisation.)
A4. altitude (Altitude (used when pressure is blank)) is taken as 0 m. (Sea-level site.)
A5. known (Second known property) is taken as relative humidity. (RH given.)
A6. process (Process) is taken as cooling with dehumidification. (Chilled-water coil.)
A7. RH_2 (Leaving relative humidity (cooling, evaporative)) is taken as 95. (Off-coil 95 % RH (typical contact factor).)

## 4. Working

```
[2] p = 101.325 (1 - 2.25577e-5 z)^5.2559                    ASHRAE Ch. 1 eq. 3
= z = 0 m
= 101.32 kPa
```
Barometric pressure (standard atmosphere)
```
[6] p_ws_1 = p_ws(t) Hyland-Wexler (entering)
= t = 28 °C
= 3.7822 kPa
```
Saturation pressure, state 1
```
[7] p_w_1 = p W/(0.621945 + W)
= W = 0.014249
= 2.2693 kPa
```
Vapour pressure, state 1
```
[8] t_1_out = dry-bulb (entering)
=
= 28 degC
```
Dry-bulb, state 1
```
[9] W_1 = 0.621945 p_w/(p - p_w)
= 2.2693 / (101.32 - 2.2693)
= 14.249
```
Humidity ratio, state 1 (g/kg dry air)
```
[10] RH_1 = p_w / p_ws
= 2.2693 / 3.7822
= 60
```
Relative humidity, state 1 (%)
```
[11] t_dp_1 = p_ws(t_dp) = p_w, bisection
= p_w = 2.2693 kPa
= 19.514 degC
```
Dew point, state 1
```
[12] t_wb_1 = adiabatic-saturation relation, bisection
= t = 28, W = 0.014249
= 22.074 degC
```
Wet-bulb, state 1
```
[13] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 28 + 0.014249 × (2501 + 1.86 × 28)
= 64.546
```
Enthalpy, state 1 (kJ/kg dry air)
```
[14] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 101.32 kPa
= 0.87267 m³/kg
```
Specific volume, state 1 (per kg dry air)

[15] rho_1 = (1 + W)/v
=
= 1.1622 kg/m³
Moist-air density, state 1
[20] p_ws_2 = p_ws(t) Hyland-Wexler (leaving)
= t = 12 °C
= 1.4026 kPa
Saturation pressure, state 2
[21] p_w_2 = p W/(0.621945 + W)
= W = 0.0082878
= 1.3325 kPa
Vapour pressure, state 2
[22] t_2_out = dry-bulb (leaving)
=
= 12 degC
Dry-bulb, state 2
[23] W_2 = 0.621945 p_w/(p - p_w)
= 1.3325 / (101.32 - 1.3325)
= 8.2878
Humidity ratio, state 2 (g/kg dry air)
[24] RH_2 = p_w / p_ws
= 1.3325 / 1.4026
= 95
Relative humidity, state 2 (%)
[25] t_dp_2 = p_ws(t_dp) = p_w, bisection
= p_w = 1.3325 kPa
= 11.225 degC
Dew point, state 2
[26] t_wb_2 = adiabatic-saturation relation, bisection
= t = 12, W = 0.0082878
= 11.553 degC
Wet-bulb, state 2
[27] h_2 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 12 + 0.0082878 × (2501 + 1.86 × 12)
= 32.985
Enthalpy, state 2 (kJ/kg dry air)
[28] v_2 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 101.32 kPa
= 0.81856 m³/kg
Specific volume, state 2 (per kg dry air)
[29] rho_2 = (1 + W)/v
=
= 1.2318 kg/m³
Moist-air density, state 2
[30] m_w = □ (W₁ - W₂)
= 2.4 × (14.249 - 8.2878) g/kg
= 0.014306 kg/s
Condensate
[31] h_w = 4.186 t₂ (condensate enthalpy)
= 4.186 × 12
= 50.232
Condensate enthalpy, kJ/kg

[32] Q_total = □ (h₁ - h₂) - □_w h_w
= 2.4 × (64.546 - 32.985) - 0.014306 × 50.232
= 75.027 kW
Total cooling load
[33] Q_sensible = □ (h(t₂, W₁) - h₂)
= 2.4 × (48.026 - 32.985)
= 36.098 kW
Sensible part (temperature change at $W_1$ ... approximated at $W_1$)
[34] Q_latent = □ (h₁ - h(t₂, W₁))
= 2.4 × (64.546 - 48.026)
= 39.648 kW
Latent part
[35] SHR = Q_sensible / (Q_sensible + Q_latent)
=
= 0.47656
Sensible heat ratio

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| t_1_out | 28 degC | Dry-bulb 1 |
| p_ws_1 | 3.7822 kPa | p_ws 1 |
| p_w_1 | 2.2693 kPa | p_w 1 |
| W_1 | 14.249 | W 1 (g/kg) |
| RH_1 | 60 | RH 1 (%) |
| t_dp_1 | 19.514 degC | Dew point 1 |
| t_wb_1 | 22.074 degC | Wet-bulb 1 |
| h_1 | 64.546 | h 1 (kJ/kg) |
| v_1 | 0.87267 m³/kg | v 1 |
| rho_1 | 1.1622 kg/m³ | ρ 1 |
| t_2_out | 12 degC | Dry-bulb 2 |
| p_ws_2 | 1.4026 kPa | p_ws 2 |
| p_w_2 | 1.3325 kPa | p_w 2 |
| W_2 | 8.2878 | W 2 (g/kg) |
| RH_2 | 95 | RH 2 (%) |
| t_dp_2 | 11.225 degC | Dew point 2 |
| t_wb_2 | 11.553 degC | Wet-bulb 2 |
| h_2 | 32.985 | h 2 (kJ/kg) |
| v_2 | 0.81856 m³/kg | v 2 |
| rho_2 | 1.2318 kg/m³ | ρ 2 |
| m_w | 0.014306 kg/s | Condensate |
| Q_total | 75.027 kW | Total cooling |
| Q_sensible | 36.098 kW | Sensible cooling |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| Q_latent | 39.648 kW | Latent cooling |
| SHR | 0.47656 | Sensible heat ratio |

## 6. Checks (governing first)

No pass/fail checks in this calculation.

## 7. Applicability

WARNING — input.h_steam: h_steam (Steam enthalpy (steam humidification)) was not entered; the template default 2676 is used.

## 8. Verification

```
Template bs.psychrometrics@1.0.0 Content hash 41c11f4a588d78b8 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```